Dominic V. Orlando

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EXPERIENCE

AI/ML Software Engineer | CSC Global, San Diego, CA — Jul 2025 – Present

- One of 2 engineers building a greenfield global obligation and risk management product, now being demoed to prospective customers
- Designed an LLM-based architecture that lets users build persistent dashboard widgets by describing the transformations and visualizations they want in natural language, cutting a custom dashboard's turnaround from ∼3 days to minutes
- Built the product's MCP and REST interfaces on Next.js and .NET, with infrastructure on Azure and Temporal
- Designed and built a self-service onboarding model that gets small- and mid-cap companies set up in minutes, instead of the weeks-long enterprise process used for Fortune 1000 clients

Founding Engineer | Terrascope, New York, NY (Remote) — Oct 2023 – Nov 2024

- On a team of 3, co-designed the product for 2 state government contracts, using satellite imagery to replace on-site visits and manual review in property tax assessment
- Engineered the ML inference pipeline, including time-series analysis, for satellite imagery across 13 counties (∼98,000 km^2), built as containerized, queue-driven workers for preprocessing, inference, and post-processing
- Built the web app's satellite map views, job dashboards, and authenticated API, plus the company website

Software Engineering Intern | Shield AI, San Diego, CA — Jun 2024 – Aug 2024

- Designed a search algorithm for a pan-tilt-zoom EO/IR payload that sweeps a track's full covariance, replacing the old approach of pointing at the track's center, which could cover most or none of it at the resolution detection algorithms require
- Implemented it in C++ and validated it with SIL and HIL testing

SELECTED WORK

AI Route Planner for Cycling and Running — TypeScript, Cloudflare, MapLibre

- Turns a natural-language description into a route using A* search over a graph created from road data
- Includes a chat interface, Strava sync, and post-ride road annotation, which lets riders mark roads they liked or want to avoid so future routes follow those notes

AI Tutoring App for Education — Next.js, LangGraph, Supabase

- Built a document ingestion pipeline that turns an instructor's course documents into text, using a vision LLM for visuals
- Grounded the agentic tutor chat and quiz generation in course material with RAG
- The client returned for a second engagement

License Plate Recognition for Parking Enforcement — React, Supabase, OpenALPR

- Led development of a system that reads plates from camera feeds, streams them live over WebSockets, and checks each one against active parking sessions
- Business outreach showed weak-to-nonexistent market signals

Strava Segment Time Predictor — TypeScript, Cloudflare D1

- Predicts segment times with a cycling power model run along the real elevation profile, then ranks winnable segments

Robot Localization, Mapping, and Planning — Python, GTSAM, OMPL

- Implemented LiDAR and visual-inertial SLAM, 3D motion planning with search- and sampling-based methods, and trajectory tracking under noise

LEADERSHIP

President | UC San Diego Cycling Team — Jun 2023 – Jun 2025

- Launched an annual race drawing 120+ unique racers, coordinating with the City of San Diego, university administration, USA Cycling, and local businesses

EDUCATION

M.S., Electrical Engineering | Machine Learning & Data Science, UC San Diego — Jun 2025

B.S., Electrical Engineering | Machine Learning & Controls, UC San Diego — Jun 2024

SKILLS

Languages	TypeScript, Python, C++, C#/.NET, SQL
Tools	Next.js, React, LangGraph, MCP, PyTorch, ROS, Azure, Temporal, Cloudflare Workers, Postgres, Docker, Kubernetes
Other	Spanish & French (conversational); BMW engine builds (turbo E46, V8-swapped E36)